Supplementary Figures

Threshold sensitivity of the ancestry-sorting analyses

Poikela et al.

The per-SNP sorting classification depends on two thresholds (Materials and Methods): the replicate-population sorting threshold τ, the minimum fraction of populations in which the dominant (unidirectional or bidirectional) component must occur for a unit to be classified as sorted; and the per-population near-fixation floor ϕ, the frequency the fixed-for (major) allele must reach for a population to count as near-fixed. We evaluated the sensitivity of the classification over a grid of $\tau \in \{0.5, 0.6, 0.7, 0.8\}$ and $\phi \in \{0.80, 0.85, 0.90, 0.95\}$. The upstream ancestry-diagnostic gate (pooled-parental minor-allele frequency ≥ 0.15) was held fixed, so the denominator—the set of differentiated SNPs—is identical across every cell of the grid and the resulting percentages are directly comparable. All other conventions match the main analysis. The primary value $\tau = 0.5$ is the logical minimum at which a sorting direction is shared by at least half of the populations.

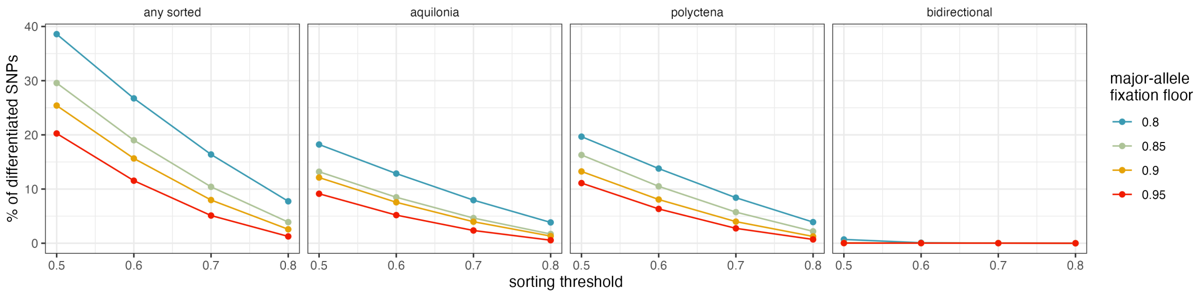

Figure S1: **Threshold sensitivity of the per-SNP sorting classification.** Percentage of differentiated SNPs assigned to each sorting outcome as a function of the sorting threshold τ (x-axis) and the major-allele near-fixation floor ϕ (line colour). The differentiated SNP set and its parental minor-allele-frequency gate were fixed across the grid. Panels show all sorted SNPs and the unidirectional *F. aquilonia*, unidirectional *F. polyctena*, and bidirectional classes. Increasing either threshold reduced the estimated prevalence of sorting monotonically, from 38.6% under the least stringent combination to 1.3% under the most stringent combination. At the primary operating point ($\tau = 0.5$, $\phi = 0.90$), 25.4% were classified as sorted. Unidirectional sorting in either ancestry direction was consistently much more common than bidirectional sorting, which approached zero as stringency increased.

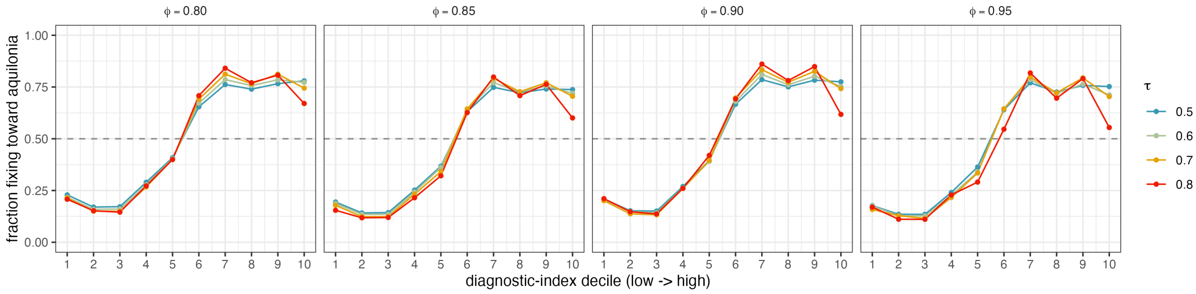

Figure S2: **Threshold sensitivity of the DI-dependent sorting direction.** Fraction of unidirectional loci fixing toward *F. aquilonia* across diagnostic-index deciles (x-axis; low to high), for each combination of the sorting threshold τ (line colour) and the major-allele near-fixation floor ϕ (panels). The dashed horizontal line at 0.5 marks no directional bias: points above it indicate a net lean toward *F. aquilonia*, points below toward *F. polyctena*. DI decile boundaries were defined once from the threshold-independent set of differentiated SNPs and held fixed across the grid. The loci contributing to each fraction nevertheless vary because changing τ or ϕ changes which loci meet the unidirectional-sorting criteria. Despite this change in the qualifying set, the low-DI-toward-*F. polyctena* / high-DI-toward-*F. aquilonia* reversal seen in main-text Figure 1B is preserved across the swept thresholds. The primary operating point ($\tau = 0.5$, $\phi = 0.90$) reproduces the main-text curve exactly.

A complementary, architectural question is whether recombination predicts the *direction* of sorting beyond the diagnostic index. Among unidirectional LD-reduced units we fit a logistic model, P(*F. aquilonia*) $\sim$ diagnostic index + recombination + parental MAF + log block size, with all predictors standardised, and asked how its coefficients respond to the sorting threshold τ (near-fixation floor held at $\phi = 0.90$). Because the per-unit sorting scores are stored, the unidirectional set was recomputed at each τ without re-running the sorting test, and the standardisation was fixed over the full differentiated-unit set, so the coefficients move only because τ changes which units qualify.

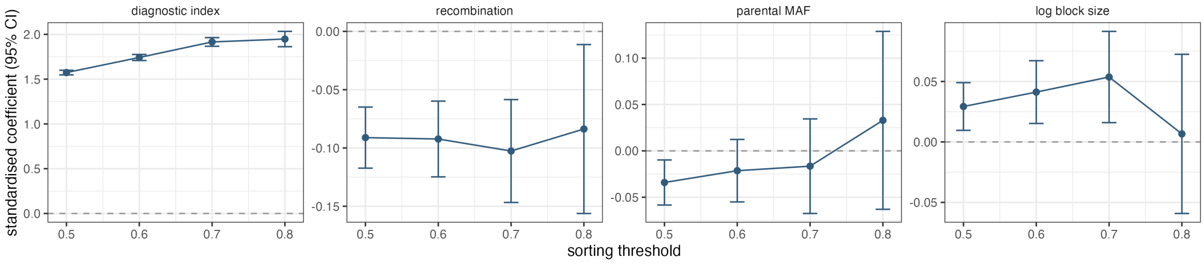

Figure S3: **Sorting direction is governed by the diagnostic index across sorting thresholds.** Standardised coefficients (points; 95% confidence intervals) of the logistic direction model P(*F. aquilonia*) ∼ diagnostic index + recombination + parental MAF + log block size, fitted to unidirectional LD-reduced units, as a function of the sorting threshold τ (near-fixation floor $\phi = 0.90$). Panels, left to right: diagnostic index, recombination, parental minor-allele frequency and log block size; the dashed line marks a null coefficient. The diagnostic index dominates at every threshold and strengthens as τ increases (from $+1.57$ at $\tau = 0.5$ to $+1.95$ at $\tau = 0.8$), whereas recombination contributes only a small, stable negative coefficient (≈ -0.09; low recombination leans *F. aquilonia* once the diagnostic index, parental MAF and block size are controlled) that remains roughly an order of magnitude weaker than the diagnostic index throughout, losing precision only at the strictest threshold as the unidirectional set shrinks. Parental MAF and block size stay near zero.

Finally, the sorting signal has clear genomic structure that persists across the sorting threshold. Figure S4 maps, along each chromosome, the fraction of differentiated SNPs fixing toward each parent in 100-kb windows at $\tau \in \{0.5, 0.6, 0.7, 0.8\}$ (near-fixation floor $\phi = 0.90$), together with the net direction, the diagnostic-index landscape and the marker-level local-LD statistic $\mathrm{ld}_w(0.95)$. Windows with fewer than 50 differentiated SNPs are removed as low-density, since their per-window estimates are unstable.

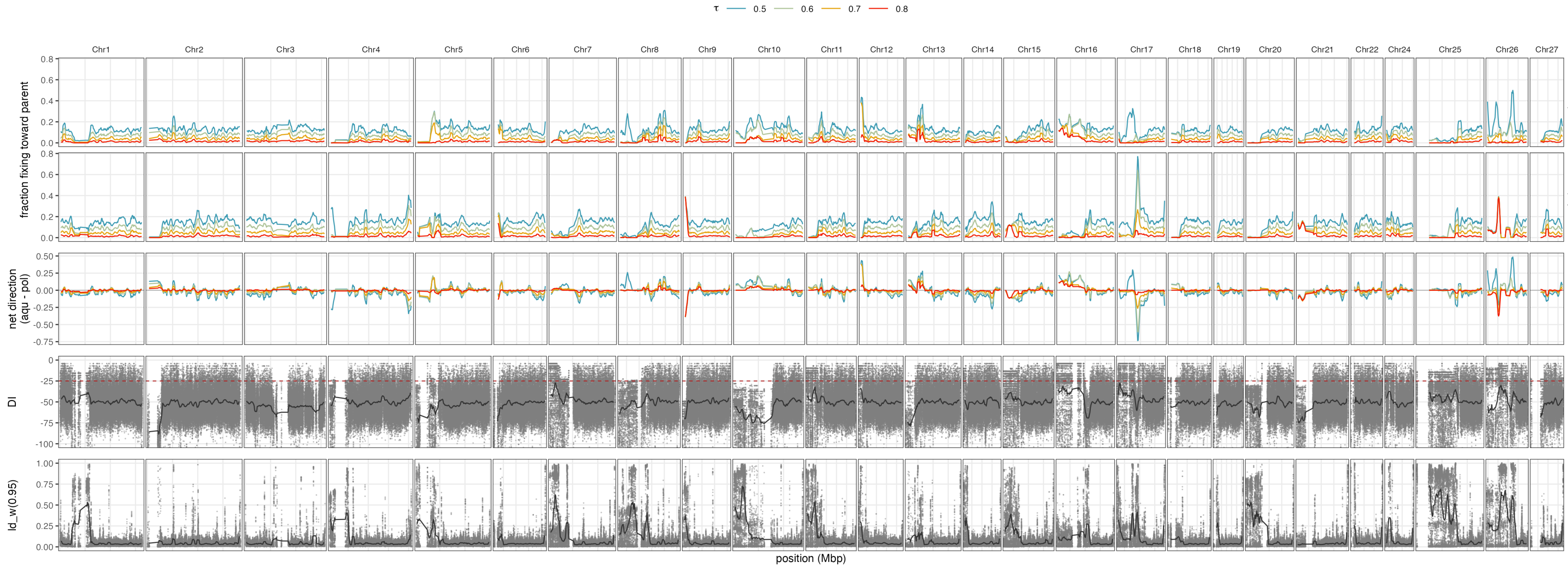

Figure S4: **Genomic landscape of directional ancestry sorting across sorting thresholds.** For each 100-kb window along the genome (faceted by chromosome), from top to bottom: the fraction of differentiated SNPs fixing toward *F. aquilonia*, the fraction fixing toward *F. polyctena*, the net direction (*F. aquilonia* minus *F. polyctena*), the diagnostic-index landscape (grey points are individual markers, line is the 5-window rolling mean, view clipped to $[-100, 0]$, dashed line marks the $\mathrm{DI} = -25$ diagnostic cut), and the marker-level local-LD statistic $\mathrm{ld}_w(0.95)$ (grey points, line is the window mean). Sorting rows are coloured by sorting threshold τ (near-fixation floor $\phi = 0.90$; 5-window rolling mean). Windows with fewer than 50 differentiated SNPs are removed as low-density. Sorting peaks frequently overlapped regions of elevated local LD, whereas the diagnostic index primarily predicted the direction of sorting. Both patterns persisted at more stringent sorting thresholds.